

Blockchain Interoperability Landscape

Inwon Kang, Aparna Gupta, Oshani Seneviratne
Presented in IEEE BigData'22



Before we start

- Much of the content is very out of date.

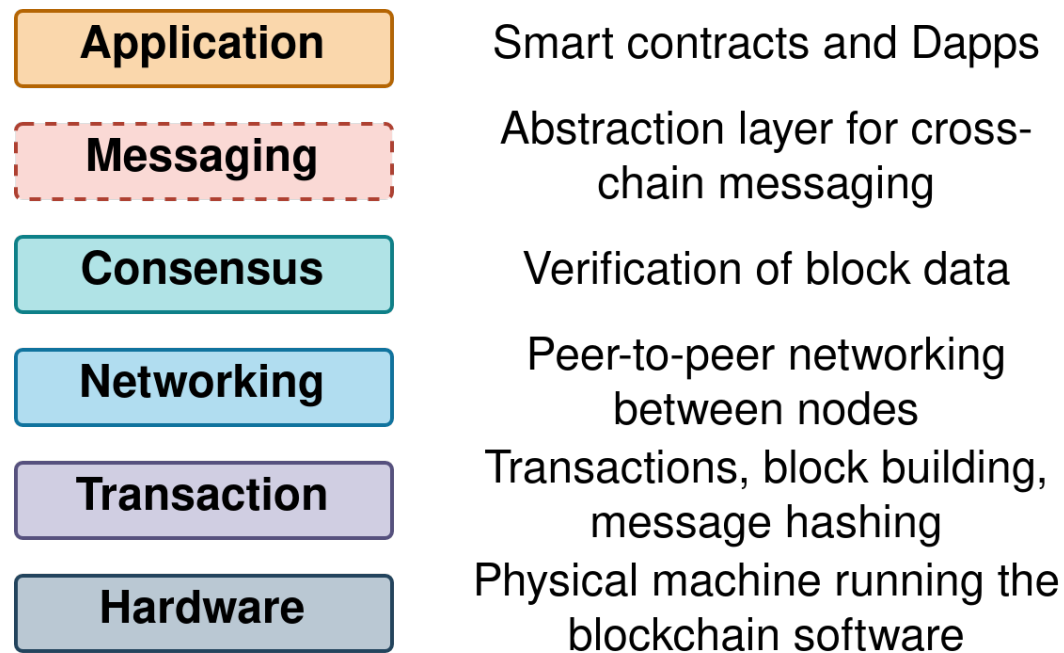
What is Blockchain Interoperability?

- NIST definition: “a composition of distinguishable blockchain systems, each representing a unique distributed data ledger, where atomic transaction execution may span multiple heterogeneous blockchain systems, and where data recorded in one blockchain are reachable, verifiable, and referable by another foreign transaction in a semantically compatible manner”
- **Ability for a given blockchain to join multiple blockchain networks and other software systems in a scalable manner**
- A blockchain interoperability solution must enable a blockchain to conduct transactions utilizing information housed in another information system in a verifiable manner

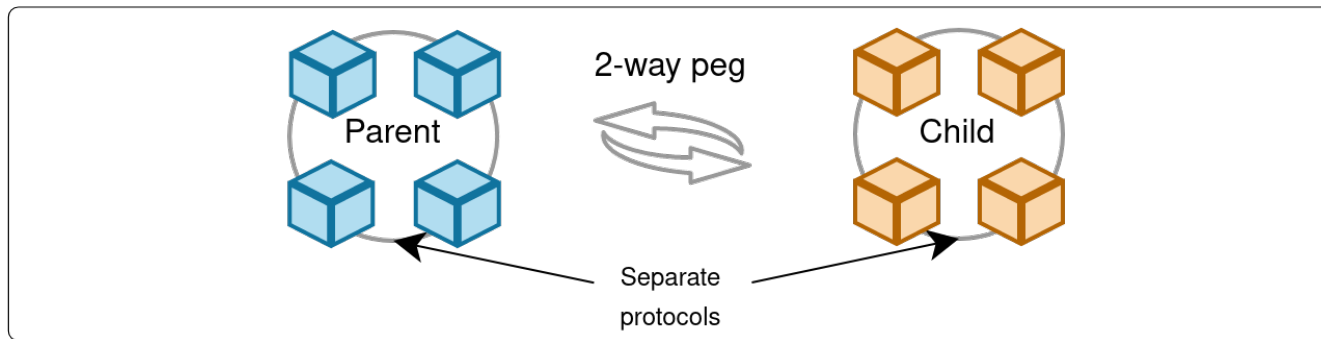
Why Blockchain Interoperability?

- Asset Movement
- Web3 Applications
- Secure and Accountable Information Exchange

Layers of Blockchain Interoperability

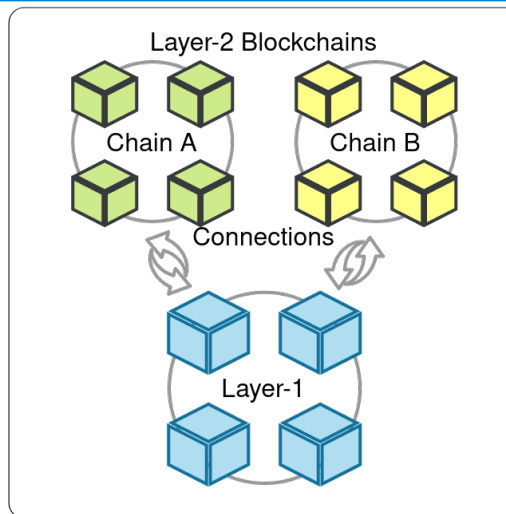


Variations in Implementations – Sidechain



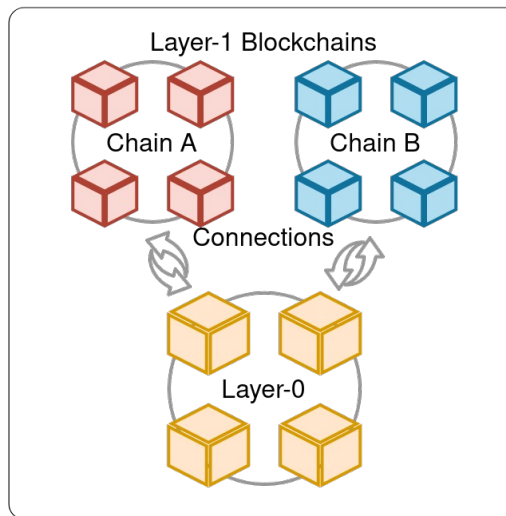
- A separate blockchain and can be connected to the parent chain to allow asset movement.
- Two-way peg mechanism makes the asset movement possible, which enables the chain-specific asset to be wrapped into the supported asset on the destination chain.
- Characterized by the fact that they are connected to their parent chains and run parallel to them.

Variations in Implementations – Layer 2



- Similar to Sidechains
- Sidechains maintains their state/protocol separately, layer-2 chain extends the functionality of the layer 1 and depends on it

Variations in Implementations – Layer 0



- Acts as a layer 1 for other layer 1 blockchains. Similar mechanism to layer 2

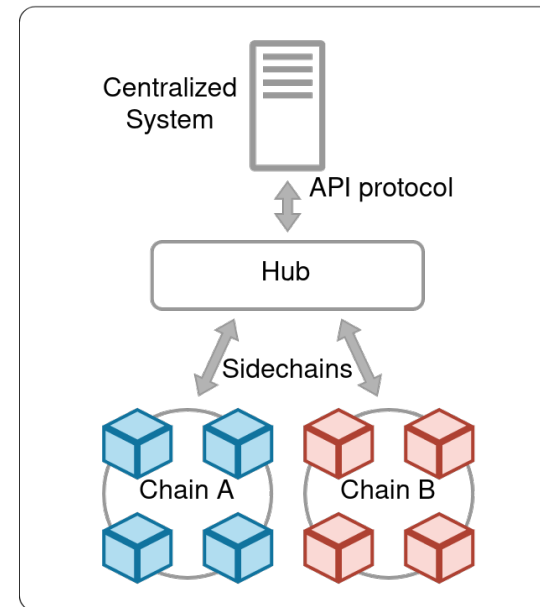
Characterization of Solutions

Overledger Provides a <i>chain-agnostic</i> 'messaging layer' for blockchain-based applications.	Polkadot Layer-0 solution for blockchain interoperability. Uses Parachains and Relay chain.	Hybrix Peer-to-peer protocol to achieve interoperability.
Hyperledger Cactus Enterprise-grade software for cross-chain transactions	Cosmos Layer-0 solution. Uses Zones and Hubs.	Chainlink Network of <i>Decentralized Oracle Networks (DONs)</i> . Interoperability through data sharing.
Hyperledger Firefly API abstraction layer for applications based on the blockchain	Loom Ethereum sidechain using LOOM token. Connects to different chains using gateways.	Poly Network Provides protocol for interoperating consortiums.

Characterization: Integration Mode

Hub and Spoke

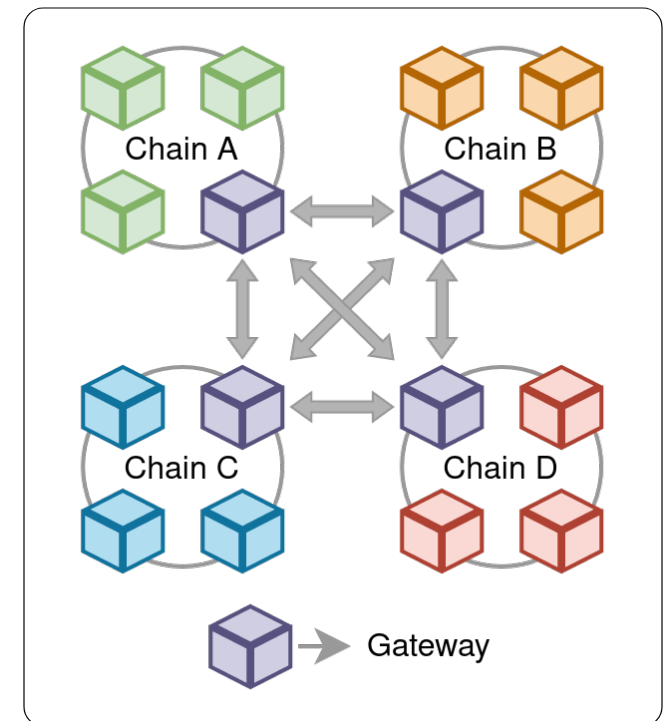
- Polkadot
- Cosmos
- Loom
- Poly Network



Characterization: Integration Mode

Decentralized Relay

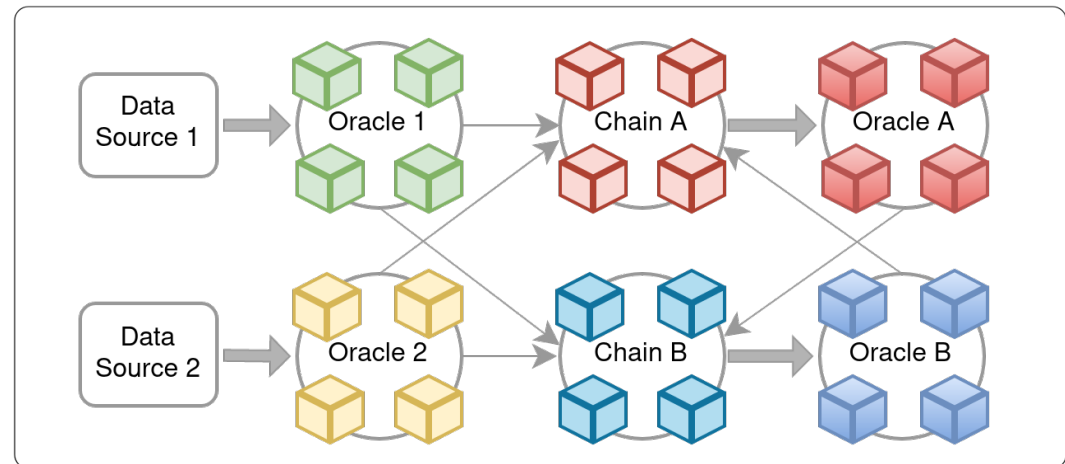
- Overledger
- Hyperledger Cactus
- Hyperledger Firefly
- Hybrix



Characterization: Integration Mode

Decentralized Oracle

- Chainlink



Characterization: Validation

Native Validation

- Validation happens *inside* the interoperability solution.

Polkadot Relay chain state is maintained by validator nodes.	Cosmos Each zone updates its parent hub with the state of its connected blockchain.
Loom Using Delegated Proof of Stake (DPoS), randomly chosen validators validate.	Chainlink Randomly chosen nodes of each <i>DON</i> validates results before sending.

Characterization: Validation

Third Party Validation

- Task of validating is delegated to a third party who is not a part of the interoperability solution.

Hyperledger Cactus

Validation is delegated to trusted party(ies) through *Ledger Plugin*.

Hyperledger Firefly

Validation is delegated to trusted party(ies) through *Blockchain Plugin*.

Characterization: Validation

Hybrid Validation

- Task of validation is partially done on the solution.

Overledger

Uses *TrustTag* to generate and verify hashes using the tx's 'comment' field.

Hybrix

Initiates *recipe* to establish protocol and validation process.

Poly Network

Chains can validate each other through Poly Chain's synced headers.

Characterization: Upgradability

Software Based

- Solutions that are based on client running a software

Overledger Connector service can be upgraded without breaking system.	Hyperledger Cactus Connector service can be upgraded without breaking system.
Hyperledger Firefly Connector service can be upgraded without breaking system.	Hybrix Connector service can be upgraded without breaking system.

Characterization: Upgradability

Blockchain Based

- Solutions that use a new blockchain to achieve interoperability

Polkadot Has <i>meta-protocol</i> , allowing protocol to change through stakeholder votes.	Cosmos May face hard fork.	
Loom May face hard fork.	Chainlink Has <i>contract-update</i> service, allowing smart contract to update itself.	Poly Network May face hard fork.

Characterization: Adoptability

Hub and Spoke

- Complex architectures can lead to more requirement for networks to join.

Polkadot Existing chains may need to implement own parachains/bridge.	Cosmos More lenient compared to Polkadot, but custom zone needs to be built.
Loom Participating chain must support ERC-20 tokens.	Poly Network Network must support light verifications, such as SPV or LES

Characterization: Adoptability

Decentralized Relay

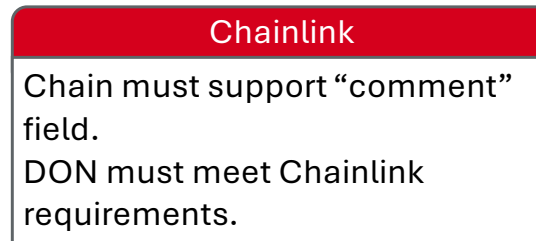
- Interoperability happens through a meta field that can contain any data structure.

Overledger Chain must support “comment” field. Chain must be publicly readable.	Hyperledger Cactus Chain must support “comment” field. Chain must be publicly readable.
Hyperledger Firefly Chain must support “comment” field. Chain must be publicly readable.	Hybrix Chain must support “comment” field. Chain must be publicly readable.

Characterization: Adoptability

Decentralized Oracle

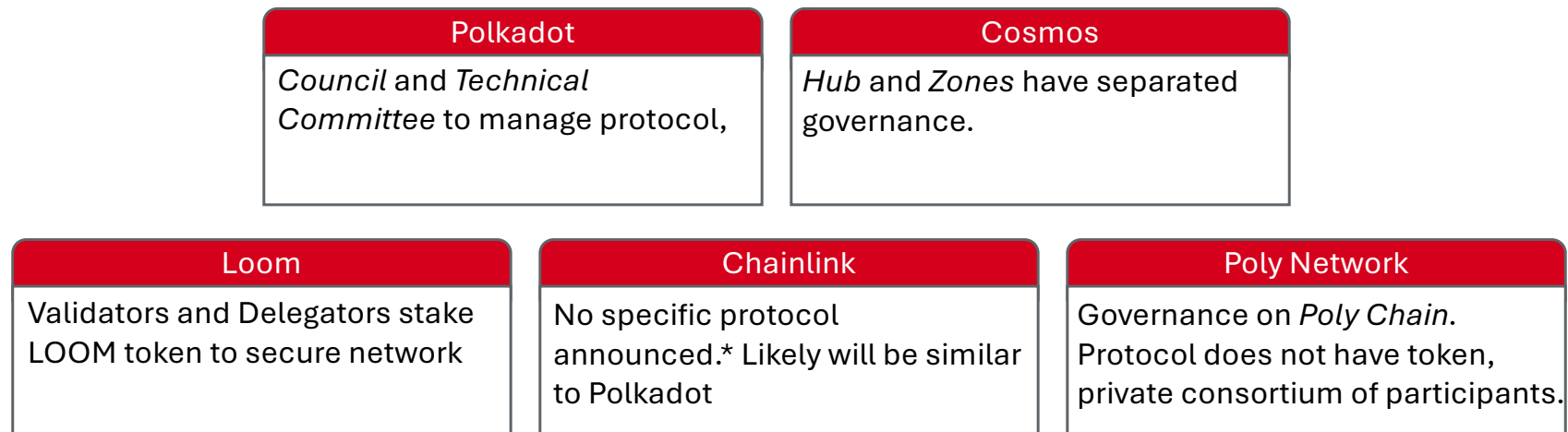
- Interoperability happens through the oracles' *data sources*.



Characterization: Governance

On-Chain

- Governance usually happens on the *main chain*.



Characterization: Governance

Off-Chain

- Usually targeted for private consortiums, no explicit governance.

Overledger QNT tokens required for software license. Consortium will have own governance.	Hyperledger Cacti Private consortium governance.
Hyperledger Firefly Private consortium governance.	Hybrix Private consortium governance.

Characterization: Trust

Overledger	Polkadot	Hybrix
Trusted	Trustless	Trusted
Hyperledger Cactus	Cosmos	Chainlink
Trusted	Trustless	Trustless
Hyperledger Firefly	Loom	Poly Network
Trusted	Trustless	Trustless

Characterization: Open Source

Overledger Proprietary	Polkadot Open Source	Hybrix Open Source
Hyperledger Cactus Open Source	Cosmos Open Source	Chainlink Open Source
Hyperledger Firefly Open Source	Loom Open Source	Poly Network Open Source

Summary of solutions

Hub and Spoke

- Higher Security.
- May be more difficult for existing chains to join.
- May be difficult to implement changes.

Decentralized Relay

- Lower Security.
- Plug-and-play software for chains to utilize.
- Better fit for smaller consortiums.

Decentralized Oracle

- Novel solution to interoperability
- May be difficult to implement complex message passing
- Can easily join off-chain and on-chain data

Decyphering Crypto Twitter

Inwon Kang, Maruf Ahmed Mridul, Abraham Sanders, Yao Ma,
Thilanka Munasinghe, Aparna Gupta, Oshani Seneviratne
Presented in WebSci'24



The Rise of Cryptocurrency

- Bitcoin was released in 2009.
- Ethereum was released in 2015.
- ICO (Initial Coin Offering) craze of 2017.
- \$2.32 Trillion market cap by 2017.

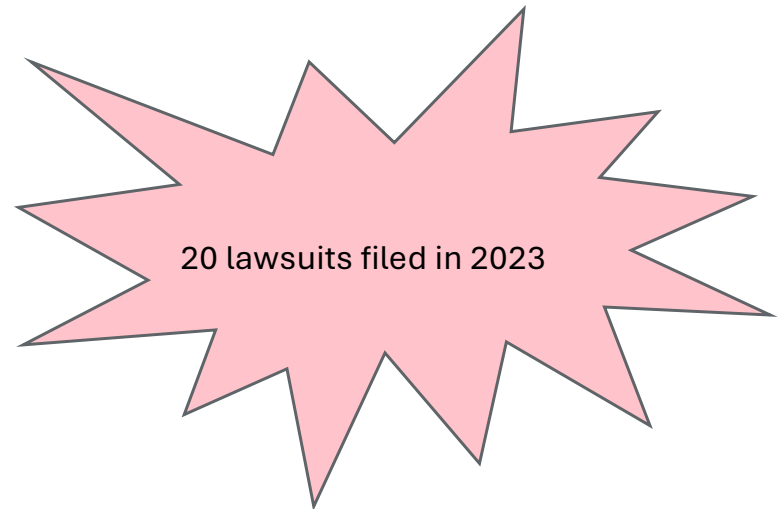


The Rise of Cryptocurrency

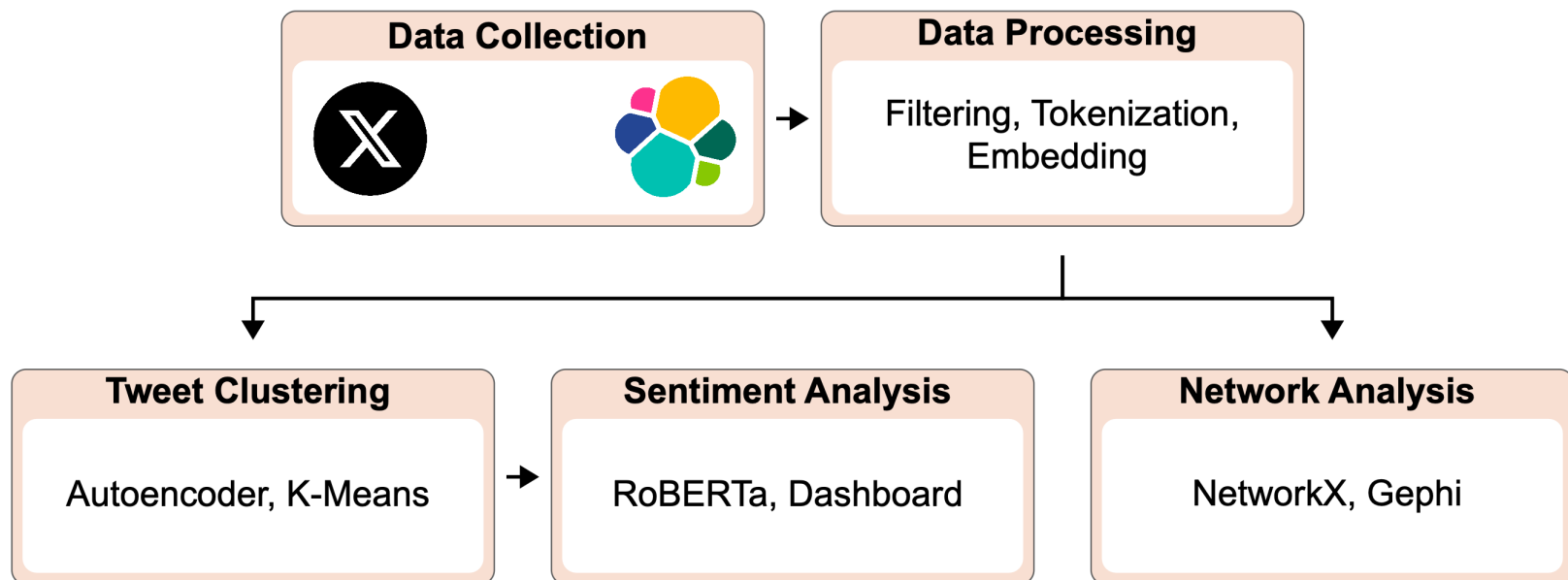
But all was not well...

The Rise of Cryptocurrency

- 2022, 2023 were rough years for crypto



Methodology



Cluster Identification

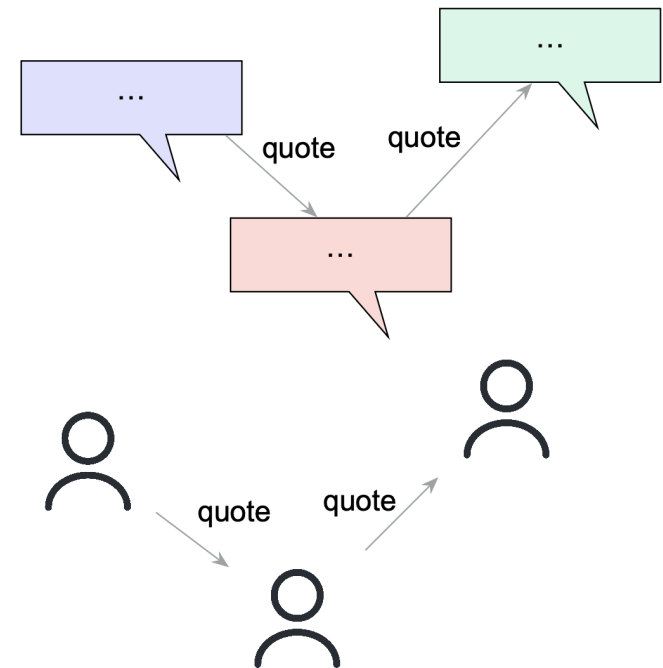
- BERT embeddings reduced to 10 dimensions using autoencoder.
 - Allows for batched training – does not require loading all data to memory at once.
- K-Means with elbow method – identified 6 clusters.
- Streamlit dashboard for visualization & exploration of clusters.



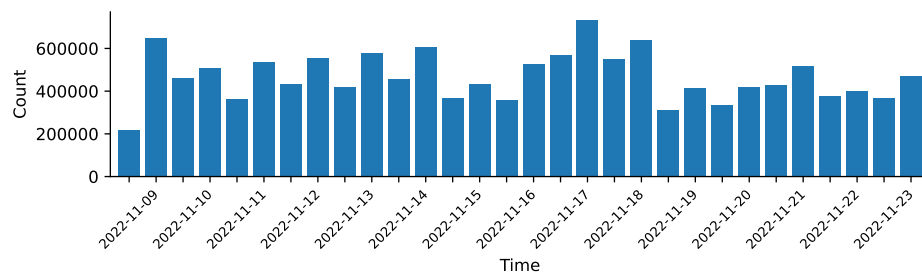
Network Construction

Graph Type	# Nodes	# Edges
Tweet-Quote	44,300	26,255
Tweet-Reply	1,591,247	1,034,017
User-Quote	1,640,251	2,520,608
User-Reply	2,340,960	7,298,105

- Nodes: User, Tweet.
- Edges: Reply, Quote.



Collected Data



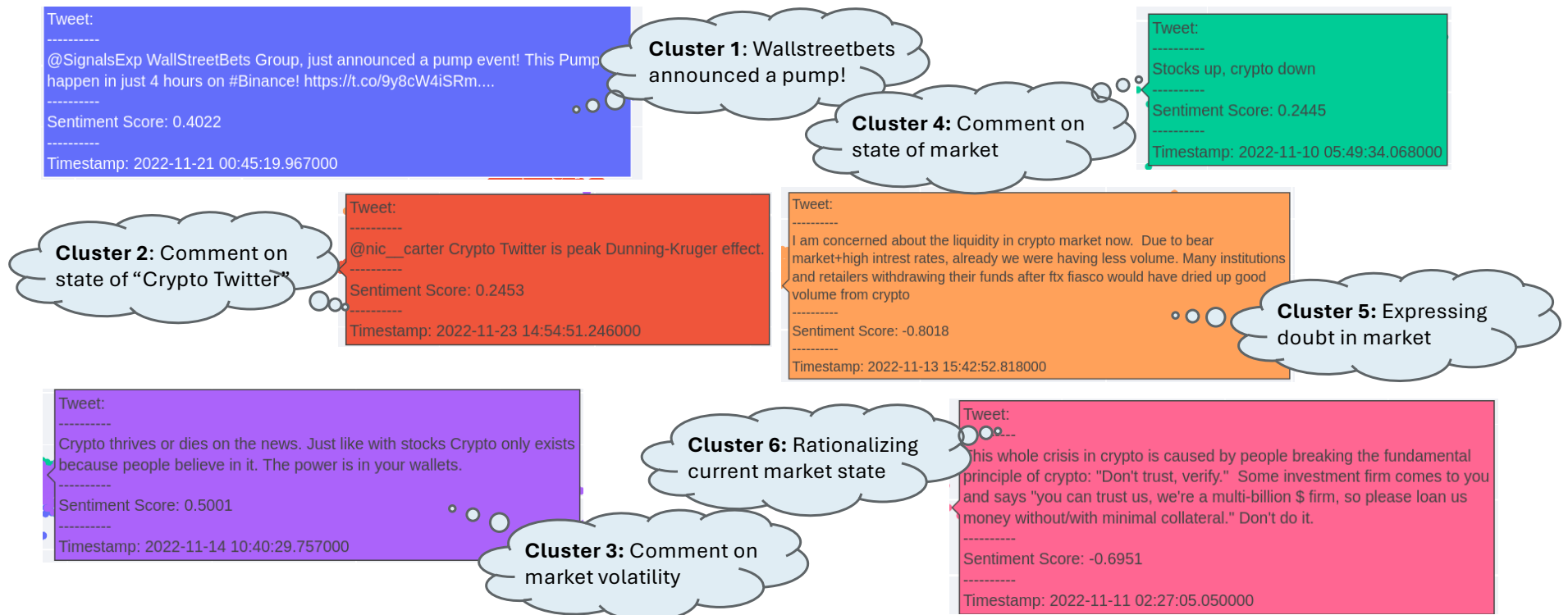
- Collection between Nov. 09, 2022 ~ Nov. 23, 2022
- Collected ~40M tweets
- ~20M after spam filtering
 - Removed retweets, repetitive spam tweets

Field	Description
id	Unique ID of tweet.
timestamp_ms	Timestamp of tweet.
user.id	User ID.
full_text	Main text of tweet.
quote_count	Number of quotes on tweet.
reply_count	number of replies on tweet.
favorite_count	Number of favorites on tweet.
quoted_status.id	ID of tweet being quoted.
quoted_status.user.id	User ID of tweet being quoted.
in_reply_to_status_id	ID of tweet being replied to.
in_reply_to_user_id	User ID of tweet being replied to.

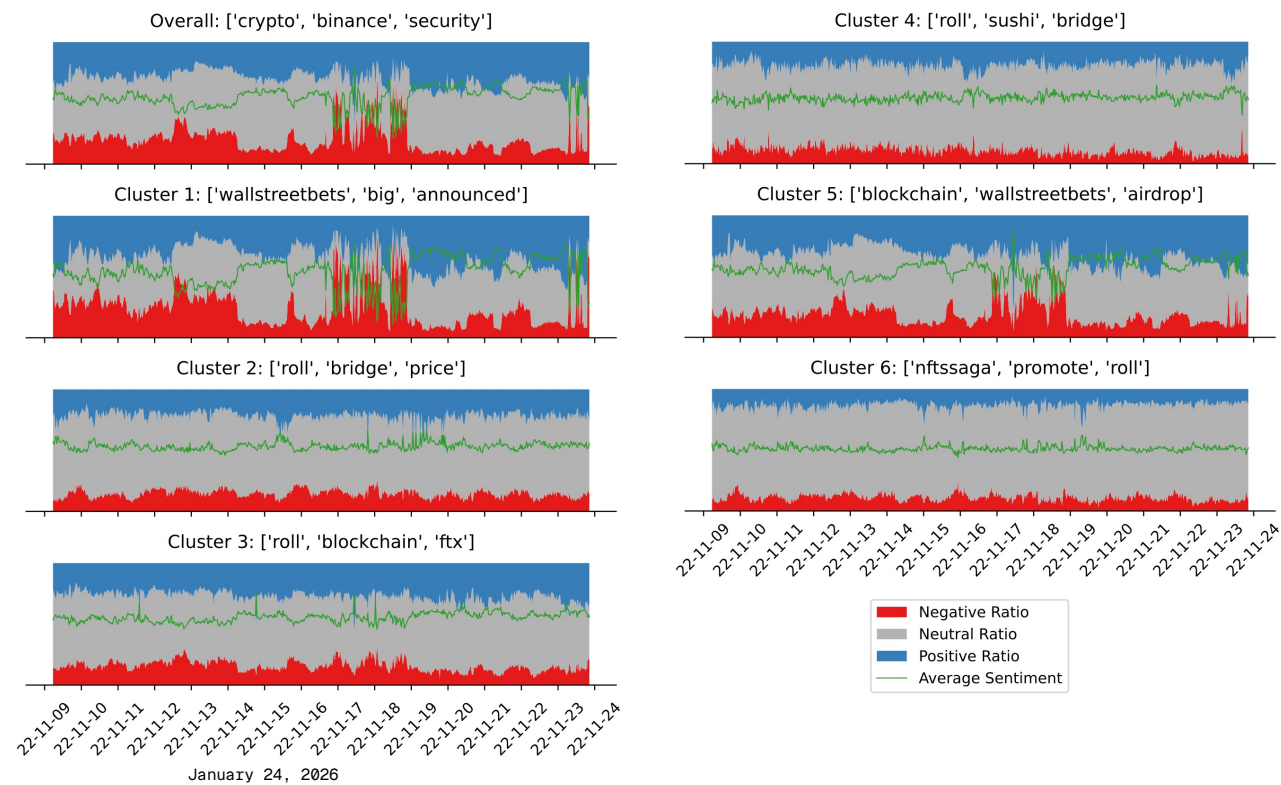
Cluster Characterization

ID	Characterization	Top 10 Keywords
	<i>Overall</i>	user, http, pump, just, signal, happen, crypto, wallstreetbets, event, kucoin
1	Crypto Conspirators	user, pump, http, signal, just, event, happen, wallstreetbets, kucoin, big
2	Meta Crypto Twitter	user, crypto, http, promote, roll, token, price, 000, binance, security
3	Crypto Observers	user, http, crypto, v2, rollup, address, tokens, claiming, compatible, evm
4	Crypto Commenters	roll, crypto, sushi, user, project, security, good, try, http, bridge
5	Crypto Doubters	user, http, pump, crypto, just, signal, kucoin, happen, event, wallstreetbets
6	Interested Investors	promote, user, crypto, price, roll, http, btc, binance, eth, bitcoin

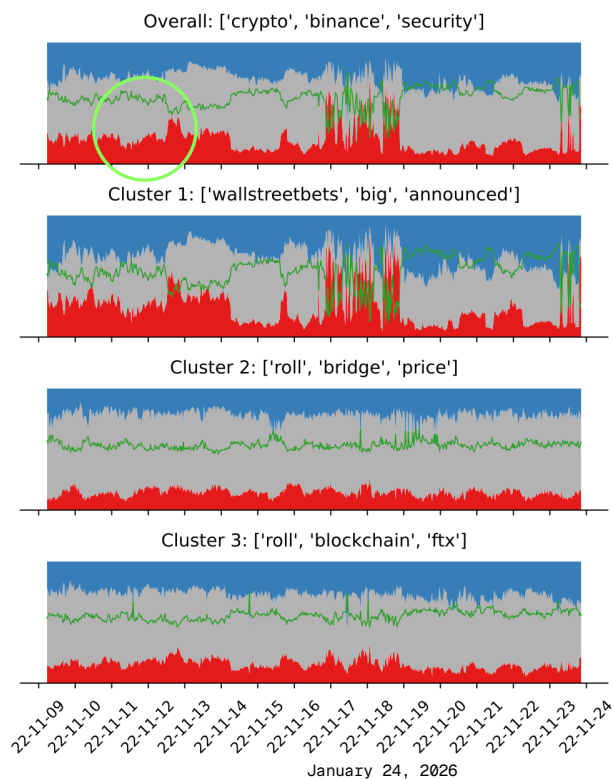
Please Read Before Editing



Sentiment Shift



Please Read Before Editing



Sam Bankman-Fried steps down as FTX CEO as his crypto exchange files for bankruptcy

PUBLISHED FRI, NOV 11 2022·9:21 AM EST UPDATED THU, MAR 28 2024·4:12 PM EDT



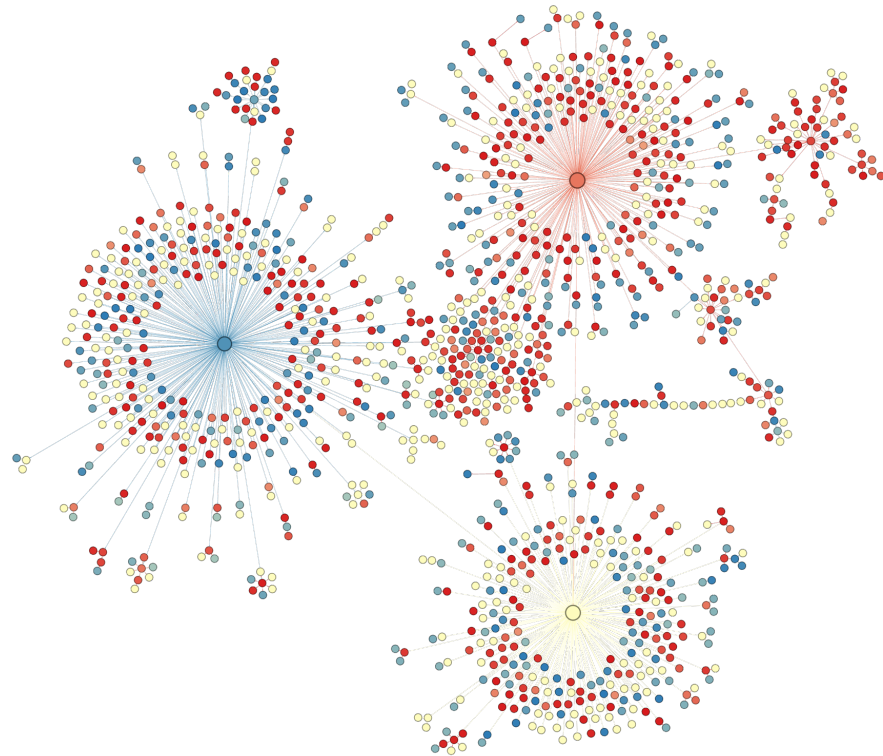
MacKenzie Sigalos
@KENZIESIGALOS

WATCH LIVE

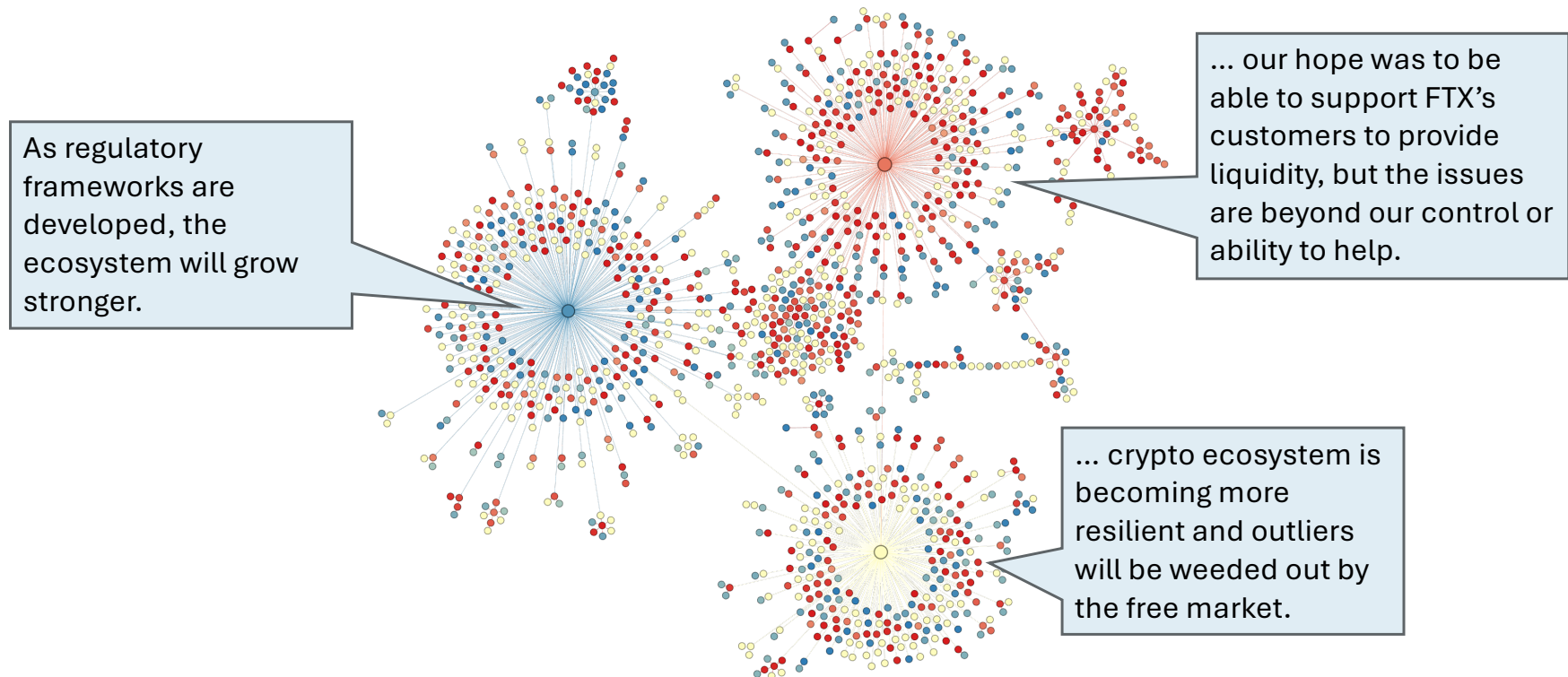
KEY POINTS

- Sam Bankman-Fried's cryptocurrency exchange FTX has filed for bankruptcy.
- Alameda Research and approximately 130 additional affiliated companies are part of the voluntary proceedings.

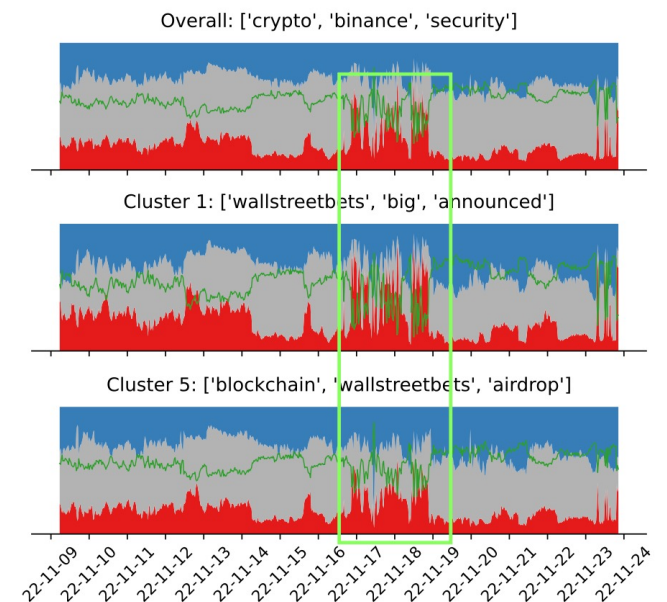
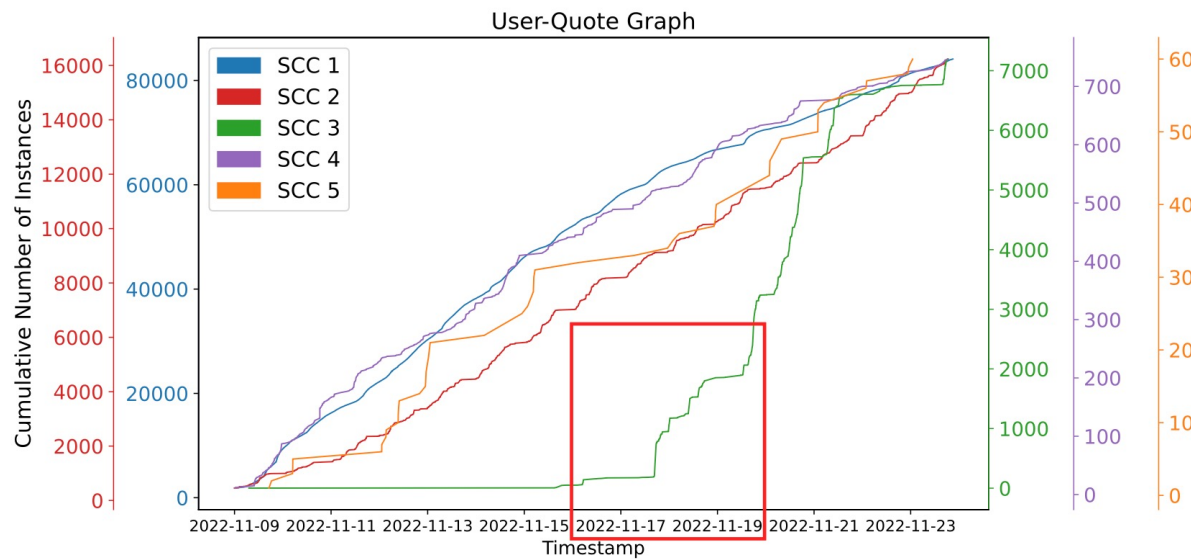
Example Tweet Network



Example Tweet Network



Example User Network



Takeaways

Good News

- Sentiment polarity can be traced back to real-life events.
 - Can social media be used to *forecast* such events?
- Communities of users with positive outlook still exist.

Bad News

- Large bot presence – many accounts that are now deleted.
- Identifiable ring of bot accounts pushing certain narratives.